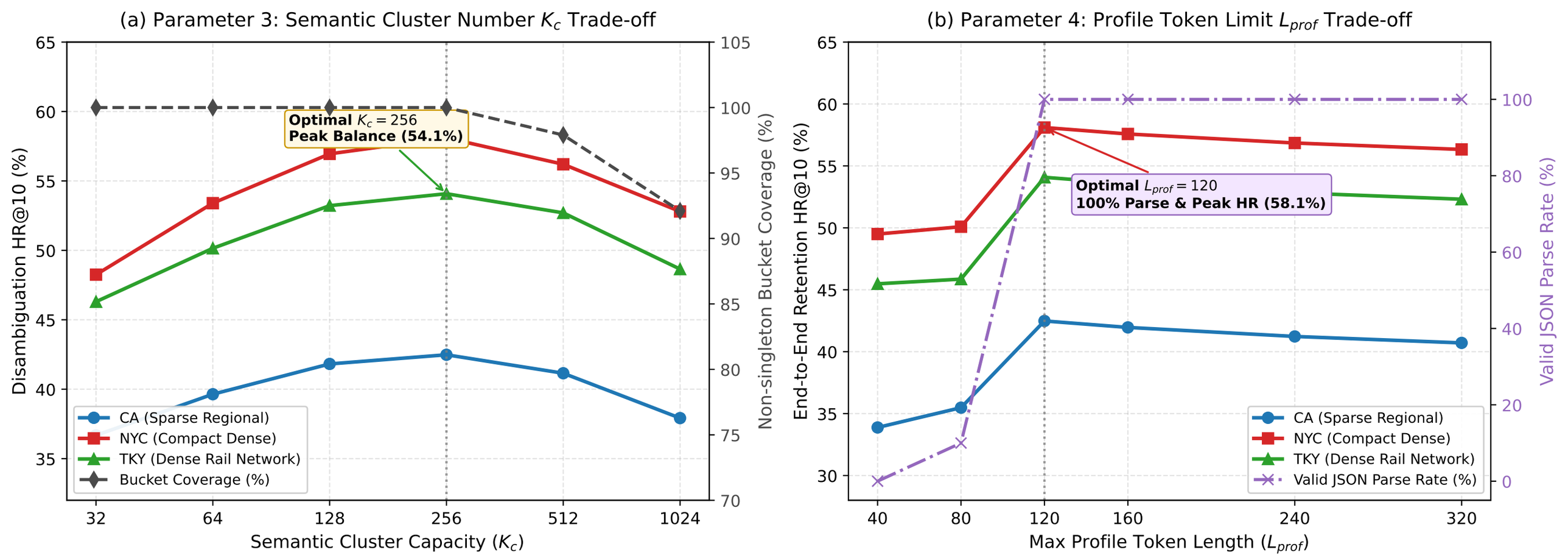
(a) Parameter 3: Semantic Cluster Number K_c Trade-off
Optimal $K_c = 256$
Peak Balance (54.1%)
Disambiguation HR@10 (%)
Non-singleton Bucket Coverage (%)
Semantic Cluster Capacity (K_c)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Rail Network)
Bucket Coverage (%)
(b) Parameter 4: Profile Token Limit L_{prof} Trade-off
Optimal $L_{prof} = 120$
100% Parse & Peak HR (58.1%)
End-to-End Retention HR@10 (%)
Valid JSON Parse Rate (%)
Max Profile Token Length (L_{prof})
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Rail Network)
Valid JSON Parse Rate (%)